A. $\mathcal{L}\{x(t-t_0)\} = e^{-st_0} \cdot X(s)$

$$\int_{-\infty}^{\infty} x(t-t_0)\, e^{-st}\, dt$$

Realizamos cambio de variable:

$u = t - t_0$ $\quad\quad$ $t \to \infty$; $u \to \infty$

$du = dt$ $\quad\quad$ $t \to -\infty$; $u \to -\infty$

$t = u + t_0$

Así: $$\int_{-\infty}^{\infty} x(u)\, e^{-s(u+t_0)}\, du$$

$$= e^{-st_0} \underbrace{\int_{-\infty}^{\infty} x(u)\, e^{-su}\, du}_{X(s)}$$

R/ Por tanto $\mathcal{L}\{x(t-t_0)\} = e^{-st_0} \cdot X(s)$

B. $\mathcal{L}\{x(at)\} = \frac{1}{|a|} X\left(\frac{s}{a}\right)$

$$\int_{-\infty}^{\infty} x(at)\, e^{-st}\, dt \; ;$$

$u = at$

$du = a\,dt$

$t = u/a$

Los límites son los mismos.

$$= \int_{-\infty}^{\infty} x(u) \cdot e^{-s\left(\frac{u}{a}\right)} \cdot \frac{1}{a}\, du$$

$$= \frac{1}{a} \int_{-\infty}^{\infty} x(u)\, e^{-\frac{s}{a}(u)}\, du = \frac{1}{a} X\left(\frac{s}{a}\right)$$

Si $a < 0$:

$$(-1)\ \frac{1}{a} \int_{-\infty}^{\infty} x(u)\, e^{-\frac{s}{a}(u)}\, du = -\frac{1}{a} X\left(\frac{s}{a}\right)$$

R/ $\mathcal{L}\{x(at)\} = \frac{1}{|a|} X(s/a)$

C. $\mathcal{L}\left\{ \frac{dx(t)}{dt} \right\} = S\, X(s)$

$$\int_{-\infty}^{\infty} \frac{dx(t)}{dt} e^{-st}\, dt$$

Utilizando la regla de la cadena:

$u = e^{-st}$ $\qquad dV = \frac{dx(t)}{dt}$

$du = -Se^{-st}dt \leftarrow V = x(t)$

$$= e^{-st} \cdot x(t) \Big|_{0}^{\infty} - \int_{-\infty}^{\infty} x(t) - Se^{-st}\, dt$$

$$[\cancel{x(\infty)} e^{\overset{0}{-\infty}} - x(0)e^{0}] + S\, \mathcal{L}\{x(t)\}$$

$x(0)$ Punto inicial

Entonces:

$$\mathcal{L}\left\{ \frac{dx(t)}{dt} \right\} = S\, X(s)$$

D $\mathcal{L}\{x(t) * y(t)\} = X(s)\,Y(s)$

$$\int_{-\infty}^{\infty} (x(t) * y(t))\, e^{-st}\, dt$$

$$= \int_{-\infty}^{\infty} x(\tau)\, y(\tau - t)\, dt$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x(\tau)\, y(\tau - t) \cdot e^{-st}\, d\tau\, dt$$

$$= \int_{-\infty}^{\infty} x(\tau)\, d\tau \int_{-\infty}^{\infty} y(\tau - t)\, e^{-st}\, dt$$

$u = \tau - t \qquad t = \tau + u$

$du = -dt$

$$= \int_{-\infty}^{\infty} x(\tau)\, d\tau \int_{-\infty}^{\infty} y(u)\, e^{-s(\tau + u)}\, dt$$

Organizando los términos

$$= \int_{-\infty}^{\infty} X(\tau) e^{-s\tau} d\tau \int_{-\infty}^{\infty} Y(u) e^{-su} du$$

$$= \mathcal{L}\{X(\tau)\} \cdot \mathcal{L}\{Y(u)\} = X(s)\,Y(s)$$

Por lo tanto $\mathcal{L}\{X(t)\,Y(t)\} = X(s)\,Y(s)$